

Version 4.1.0.6 20 May 2026

SD card wiring pin TX (GPIO 1) moved to pin 3 RX

Version 4.1.0.5 03 May 2026

LCD removed.
Sensors/Lasers moved to 5V
SD card wiring pin 17 moved to pin TX (GPIO 1)
Sound added

Version 4.0.1.0 November 2025

wiring for SD card added
wiring foTFT touch added
Optional parts wiring removed for simplicity.
Orange wires changed to blue for compatibility with connector kit.

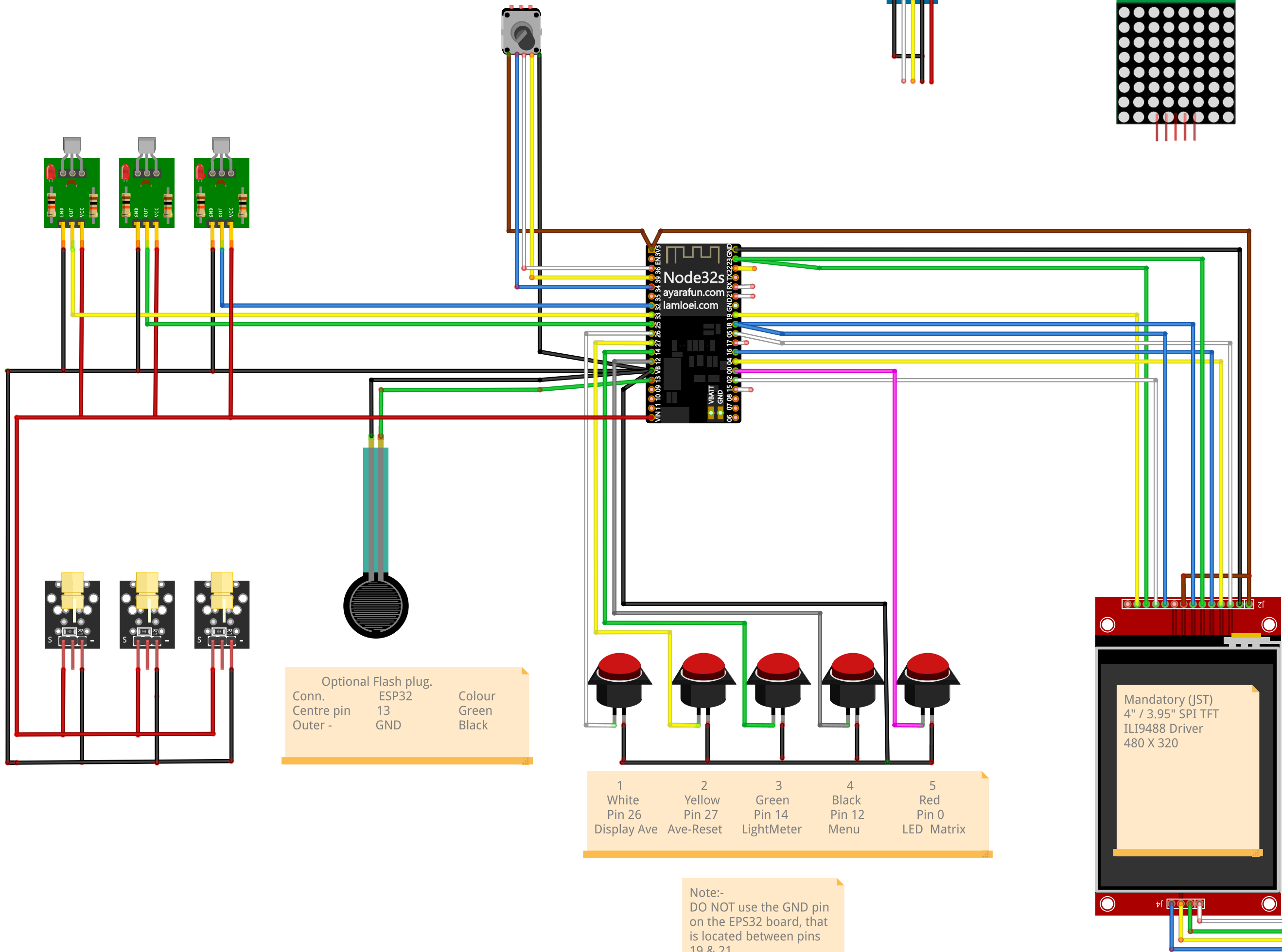
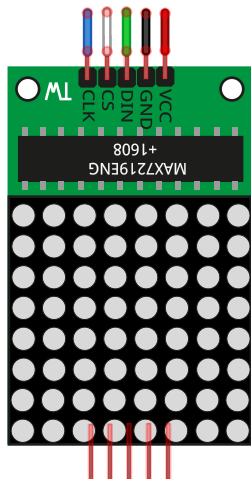
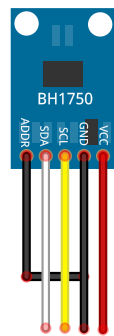
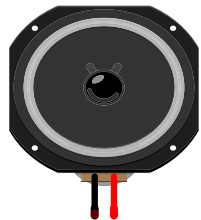
Sensors		
Sensor Out	ESP32 Pin	Wire
Colour		
sen3	33	Yellow
senM	25	Green
Sen1	32	Blue
GND	GND	Black
VCC	Vin	Red

Encoder		
Encoder	ESP32 Pin	Wire Colour
3.3V	3V3	Red
Key	34	Blue
S2	36	White
S1	39	Yellow
0V	GND	Black

OptionalLight SensorGY-302		
module	ESP32 Pin	Wire Colour
ADDR -	GND	Black
SDA	21	White
SCL	22	Yellow
GND	GND	Black
VCC	5V	Red

Optional LED Matrix		
LED	ESP32 Pin	Wire Colour
Vcc	5V	Red
GND	0V	Black
DIN	23	Green
CS	15	White

Optional Speaker		
Spkr	ESP32 Pin	Wire Colour
Red	17	White
Black	0V	Black



Caution
TFT is powered by 3V3
Ensure you correctly
wire the TFT.

Caution on TFT Wiring.
Connector is on the rear of the
TFT.
Ensure you do not wire up as a
mirror image.

Optional wiring for
SD card
3 RX/23/19/18

Note:-
DO NOT use the GND pin
on the EPS32 board, that
is located between pins
19 & 21.

TFT pins in order	ESP32	Wire Colour	
TFT_VDD	3V3	Red	
TFT_GND	GND	Black	
TFT_CS	5	White	Chip select
TFT_RST	4	Yellow	Reset pin
TFT_DC	16	Blue	Data Command
TFT_MOSI	23	Green	
TFT_SCLK	18	Blue	
TFT_LED	3V3	Brown	
TFT_MISO	----	do no connect	
Optional Touch Wiring			
Touch_CCLK	18	Blue	
Touch_CS	2	White	
Touch_DIN	23	Green	MOSI
Touch_TFT_DO	19	Yellow	MISO
Touch_IRQ		do not connect	
Optional Sd Card conections for dataLogger			
CLK (SLK)	18	Blue	
SDO (MISO)	19	Yellow	
SDI (MOSI)	23	Green	
SCS (CS)	3 RX	White	